
Vibration and Damping Analysis of a Bridge Stay Cable

Ruohan He & Yuchen Wang

12/3/2025

2D Cable Model

Geometry and Discretization Parameters

- **Number of nodes:** $n_v = 51$
- **Degrees of freedom:** $n_{\text{dof}} = 2n_v$
- **Number of elements:** $n_e = n_v - 1$
- **Cable span length:** $L = 300$ m
- **Cable inclination angle:** $\theta = 23^\circ$
- **Element length:** $\Delta l = 6$ m
- **Cable radius:** $r_0 = 0.060$ m
- **Fixed degrees of freedom:**

$$(x_0, y_0), \quad (x_{n_v-1}, y_{n_v-1})$$

Material and Stiffness Parameters (steel)

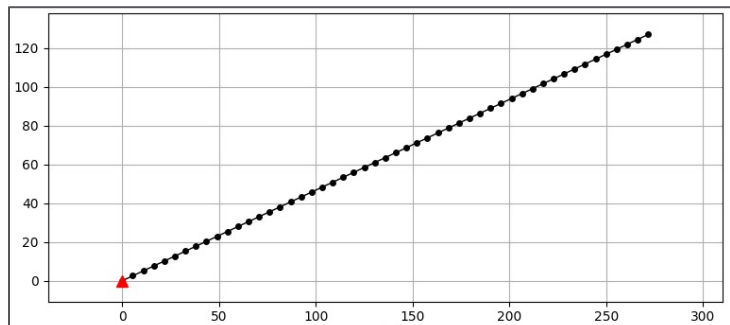
- **Young's modulus:** $Y = 200$ GPa
- **Poisson ratio:** $\nu = 0.3$
- **Material density :** $\rho = 7850$ kg/m³

Damping

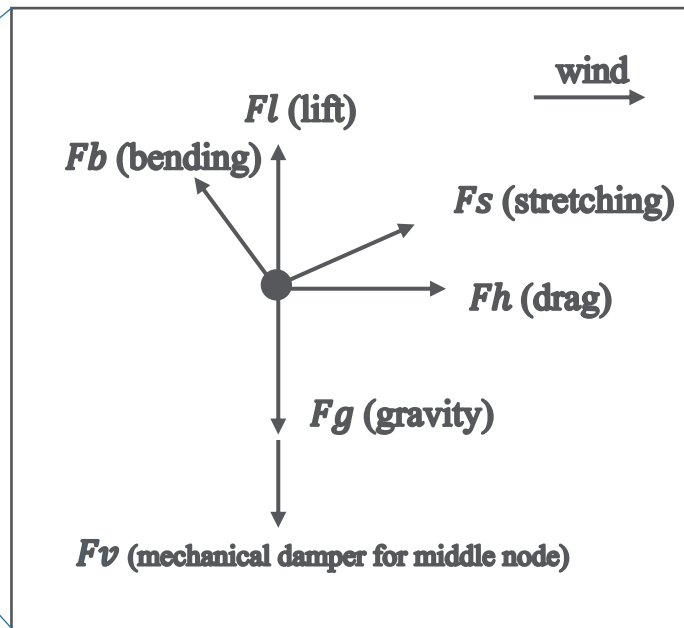
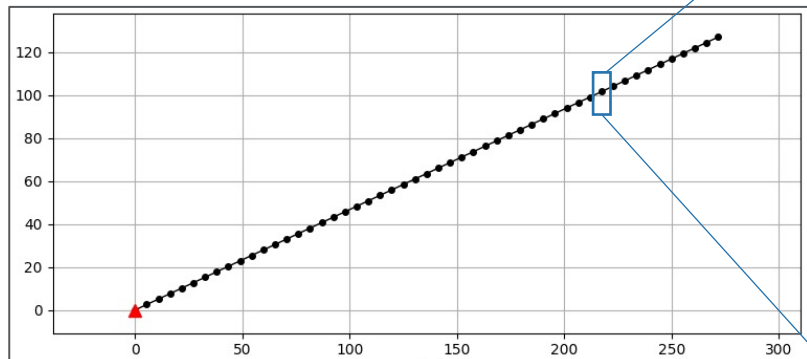
- **Mechanical damper coefficient:** $c_{\text{damp}} = 1000$ N · s/m
- **Damper location:** mid-span node $i = n_v/2$

Hydrodynamics

- **Fluid density:** $\rho_f = 1.225$ kg/m³
- **Drag coefficient:** $C_D = 0.79$
- **Lift coefficient:** $C_L = 0.3$
- **Background flow velocity:** $\mathbf{u}_f = (30.0, 0.0)$ m/s



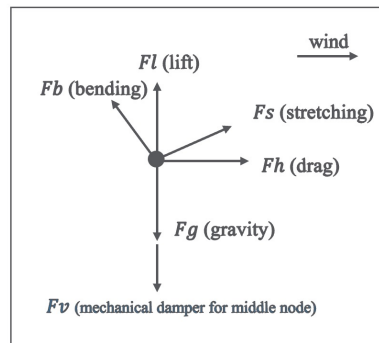
Forces Acting on a Cable Node



Forces Acting on a Cable Node

All the major forces acting on a single node of a bridge cable:

- F_s : stretching force from axial tension.
- F_b : bending resistance due to curvature change.
- F_g : gravity acting downward.
- F_h : aerodynamic drag opposite to the direction of relative wind.
- F_l : aerodynamic lift perpendicular to the flow.
- F_v : mechanical damper force used to dissipate vibrations.



$$E_s = \frac{1}{2} EA \epsilon^2 dl \implies F_s = -\frac{\partial E_s}{\partial \mathbf{x}}$$

$$E_b = \frac{1}{2} \frac{EI}{dl} (\kappa_k - \kappa_k^0)^2 \implies F_b = -\frac{\partial E_b}{\partial \mathbf{x}}$$

$$F_g = g m_i$$

$$F_h = -\frac{1}{2} \rho_f C_D A_k \|\mathbf{u}_{\text{rel}}\| \mathbf{u}_{\text{rel}}$$

$$F_v = -C \mathbf{u}$$

$$F_l = -\frac{1}{2} \rho_f C_L A_k \|\mathbf{u}_{\text{rel},k}\|^2$$

Total Energy

- **Kinetic energy:**

$$E_{\text{kin}} = \frac{1}{2} \sum_i m_i \|\mathbf{u}_i\|^2 = \frac{1}{2} \mathbf{u}^T \mathbf{M} \mathbf{u}.$$

- **Mechanical damping dissipation:**

$$P_{\text{damp}} = \mathbf{u}^T \mathbf{C} \mathbf{u}, \quad E_{\text{damp}}(t) = \int_0^t P_{\text{damp}}(\tau) d\tau.$$

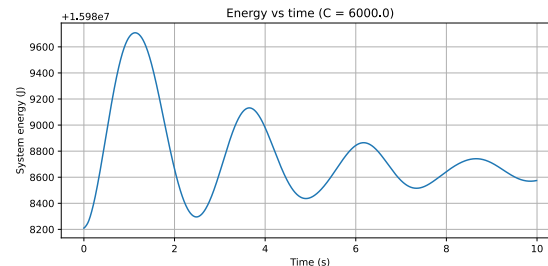
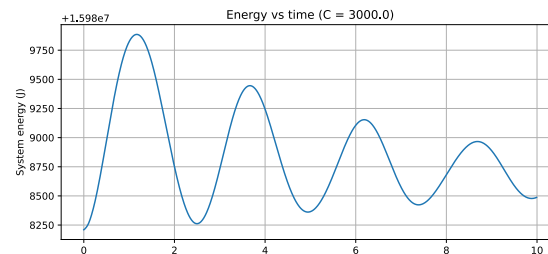
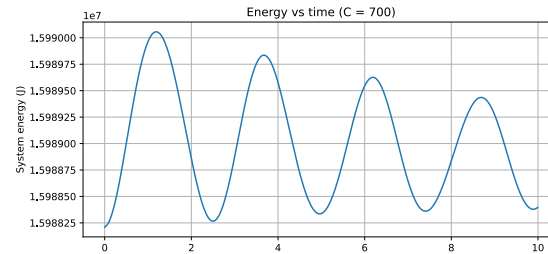
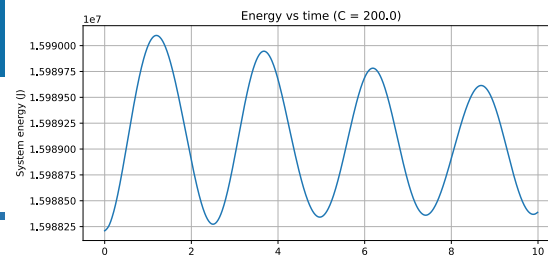
- **Aerodynamic drag dissipation:**

$$P_{\text{drag}} = -\mathbf{F}_h^T \mathbf{u}, \quad E_{\text{drag}}(t) = \int_0^t P_{\text{drag}}(\tau) d\tau.$$

- **System total energy:**

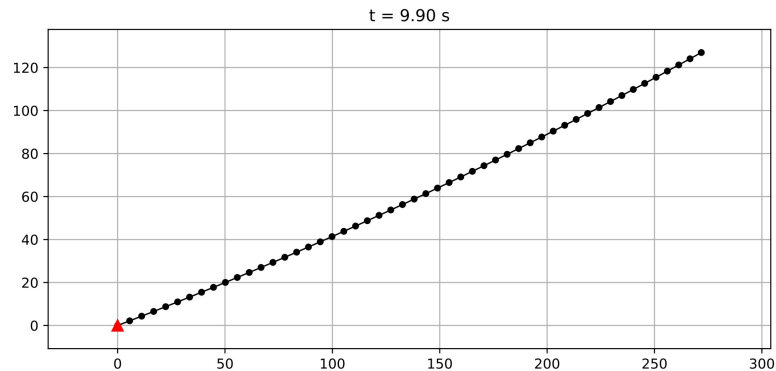
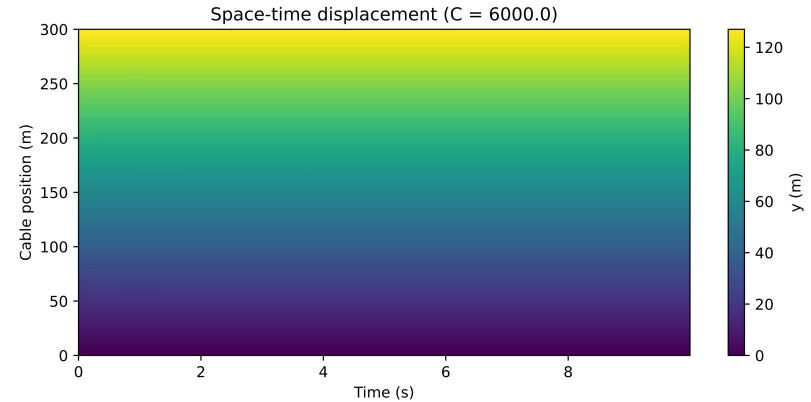
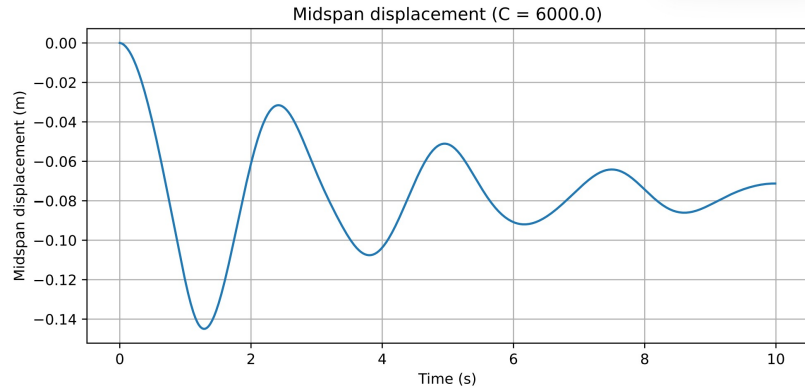
$$E_{\text{system}}(t) = E_p(t) + E_{\text{kin}}(t) - E_{\text{damp}}(t) - E_{\text{drag}}(t).$$

$$E_p = E_{\text{stretching}} + E_{\text{bending}} + E_{\text{gravity}}.$$



T=10s
dt=0.01s

Displacement & Vibration Visualization



Critical Damping

- **Natural frequency:** $\omega_n = \sqrt{\frac{k}{m}}$
- **Critical Damping:** $c_{crit} = 2m_{eff}\omega_n$
- **Damping ratio:** $\zeta = \frac{c}{c_{crit}}$
- **Recommend damper value:** $c_{recommend} = \zeta_{target}c_{crit}$

Natural frequency $f_1 = 0.40$ Hz

Natural angular frequency $\omega_1 = 2.51$ rad/s

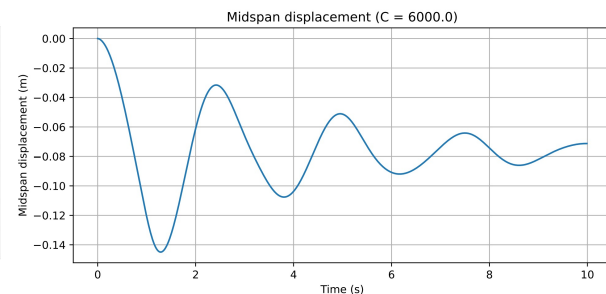
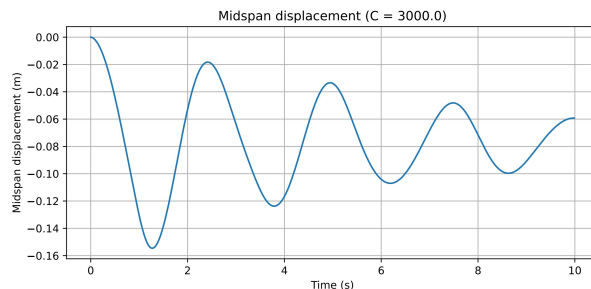
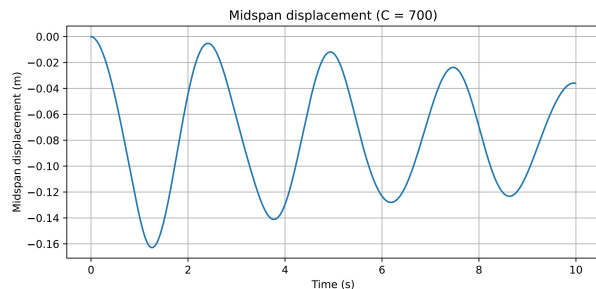
Total cable mass $M_{total} = 26634.42$ kg

Effective mass $m_{eff} = 6658.61$ kg

Target damping ratio = 0.02

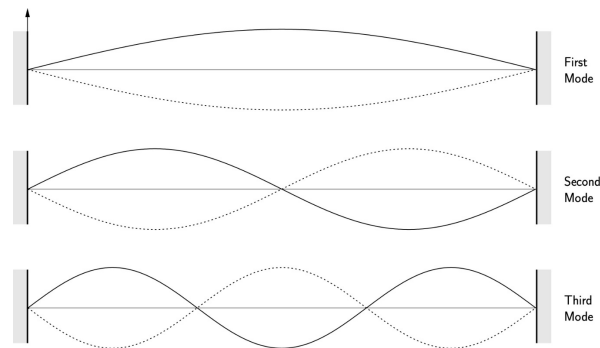
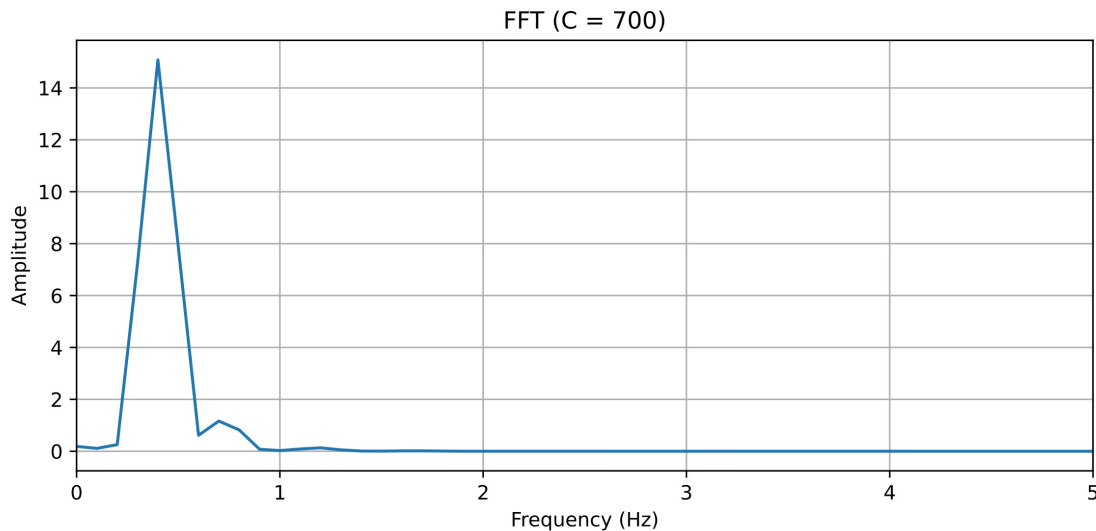
Critical damping $c_{crit} = 33469.80$ N • s/m

Recommended mechanical damper $c_{damp_new} = 669.40$ N • s/m

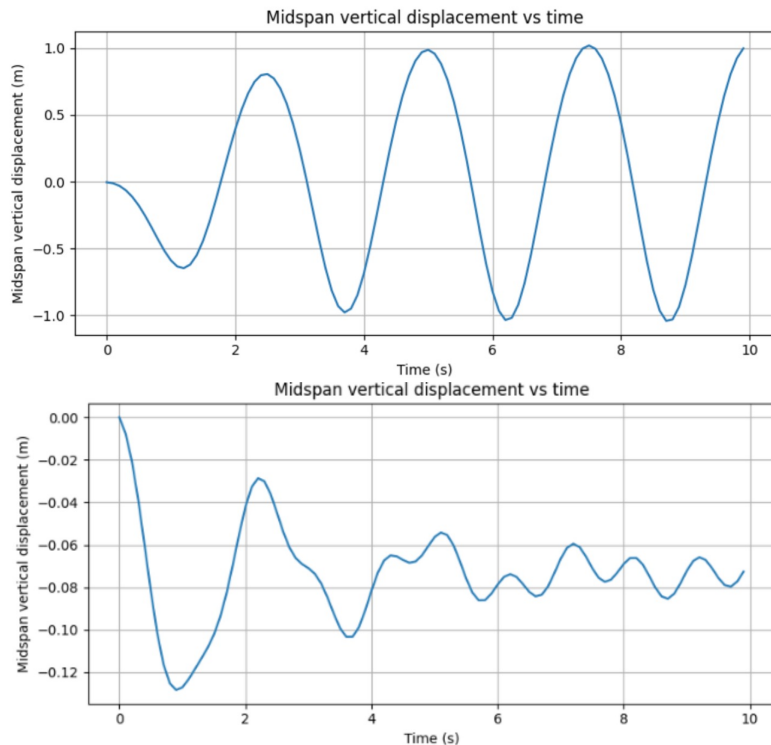


Frequency Analysis (FFT)

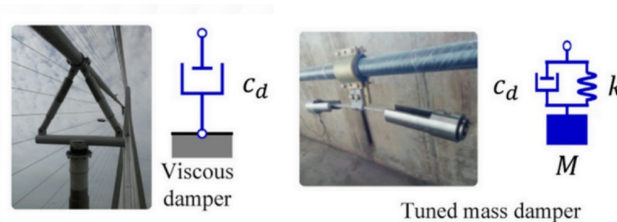
- Damped natural frequency: $\omega_d = \omega_n \sqrt{1 - \zeta^2}$



Next Steps



Adding a moving boundary at the lower anchorage would introduce parametric excitation and allow examination of support induced oscillations.



Real structures often employ multiple external dampers positioned at different locations along a stay cable, or use combined devices such as viscous dampers, tuned mass dampers.

Reference

- [1] S. Kumarasena, “Wind-Induced Vibration of Stay Cables,” FHWA- HRT-05-083, Federal Highway Administration, 2007.
- [2] Federal Highway Administration (FHWA), “Long-Term Cable Durability Study,” FHWA-HRT-17-037, U.S. Department of Transportation, 2017.
- [3] H. Jing, Y. Xia, H. Li, Y. Xu, and Y. Li, “Excitation mechanism of rain-wind-induced cable vibration in a wind tunnel,” *Journal of Fluids and Structures*, vol. 68, pp. 32–47, 2017.
- [4] PCI, “Cable-Stayed Bridges with Prestressed Concrete,” *PCI Journal*, Sept.–Oct, 1987.
- [5] E. de S’a Caetano, *Cable Vibrations in Cable-Stayed Bridges*. Zurich, Switzerland: IABSE, 2007.
- [6] L. Sun et al., “Stay cable vibration mitigation: A review,” *Structural Control and Health Monitoring*, vol. 30, no. 1, 2023, Art. e3053.
- [7] Sun, L. Chen, and H. Huang, “Stay cable vibration mitigation: A review,” *Adv. Struct. Eng.*, vol. 25, no. 16, pp. 3368–3404, Dec. 2022.
- [8] J. Yi, “Modeling and analysis of cable vibrations in cable-stayed bridges under near-fault ground motions,” *Eng. Struct.*, vol. 277, p. 115443, Feb. 2023.
- [9] P. Egger and L. Caracoglia, “New discrete mass model examining the dynamics of multiple-element-pendulum impact dampers for bridge stay-cables,” *Eng. Struct.*, vol. 253, p. 113665, Feb. 2022.
- [10] *Vibrations and Waves*, “Transverse Vibrations of a Cable,” Engineering Courses – University of Alberta.

Thank You
